

GEOGRAPHY

INDIAN GEOGRAPHY



MINERALS & INDUSTRIES

MODULE - 2

- **NUCLEAR & WIND ENERGY**
- **OFF-SHORE WIND FARMS**

INDIAN MINERAL & ENERGY RESOURCES

NON-FERROUS MINERAL

India is poorly endowed with non-ferrous metallic minerals except bauxite.

1

BAUXITE



1 Bauxite is the ore which is used in manufacturing of aluminium.

2 Bauxite is found mainly in tertiary deposits.

3

It is associated with laterite rocks occurring extensively either on the plateau or hill ranges of peninsular India and also in the coastal tracts of the country.

4

ODISHA:

Odisha happens to be the largest producer of Bauxite.

- **Kalahandi and Sambalpur** are the leading producers.
- The other two areas which have been increasing their production are **Bolangir and Koraput**.

INDIAN MINERAL & ENERGY RESOURCES

NON-FERROUS MINERAL

1

BAUXITE

5 Gujarat, Chhattisgarh, Madhya Pradesh and Maharashtra are other major producers.

6 GUJARAT:
Bhavnagar, Jamnagar in Gujarat have the major deposits.

7 CHHATTISGARH: It has bauxite deposits in Amarkantak plateau.

8 MADHYA PRADESH: Katni-Jabalpur area and Balaghat in M.P. have important deposits of bauxite.

9 JHARKHAND:
The pat lands of Jharkhand in Lohardaga have rich deposits.

10 MAHARASHTRA:
Kolaba, Thane, Ratnagiri, Satara, Pune and Kolhapur are the important producers here.

11 Tamil Nadu, Karnataka and Goa are minor producers of bauxite.

NON-FERROUS MINERAL

2

COPPER



1 Copper is an indispensable metal in the electrical industry for making wires, electric motors, transformers and generators.

2 It is alloy able, malleable and ductile. It is also mixed with gold to provide strength to jewellery.

3

The Copper deposits mainly occur

- in **Singhbhum district** in Jharkhand,
- **Balaghat district** in Madhya Pradesh and
- **Jhunjhunu and Alwar** districts in Rajasthan.

4

Minor producers of Copper are

- Agnigundala in Guntur District (Andhra Pradesh),
- Chitradurg and Hasan districts (Karnataka) and
- South Arcot district (Tamil Nadu).

NON-FERROUS MINERAL

3

GOLD

1 Gold usually occurs in **auriferous** [of rocks or minerals containing gold] rocks.

2 It is also found in **sands** of several rivers.



KOLAR GOLD FIELDS

3

KARNATAKA:

- Karnataka is the largest producer of gold in India.
- Gold mines are located in Kolar [**Kolar Gold Field**], Dharwad, Hassan and Raichur [**Hutti Gold Field**] districts.
- Kolar Gold Fields is one of the deepest mines of the world.
- **Hutti mines** are exploited to their maximum levels and the ore left behind is of very low grade. The mining has almost ceased due to little or no profitability.
- The Kolar Gold Field has also run out of quality reserves.

NON-FERROUS MINERAL

3

GOLD

4 ANDHRA PRADESH:

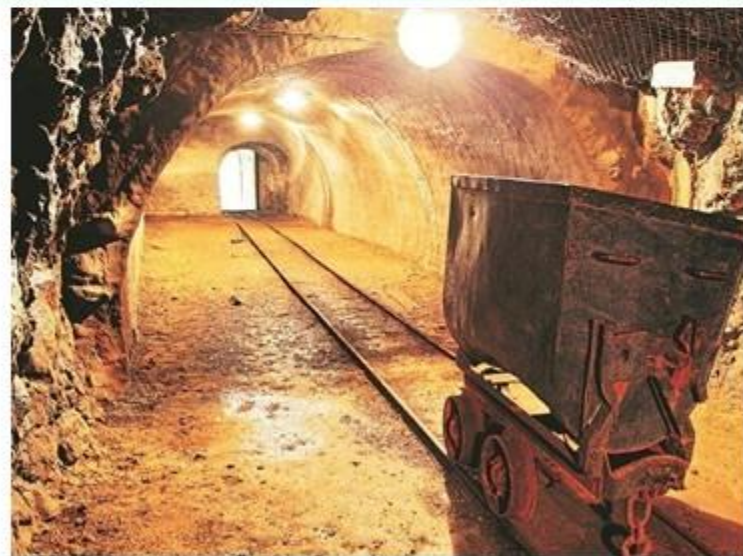
- Second largest producer of gold in India.
- Ramagiri in Anantapur district is the most important gold field in AP.

5 JHARKHAND:

- Alluvial Gold [gold scattered in silt] and Placer deposits [gold bearing rocks] in small quantity are widely spread in a large number of rivers
- Sands of the Subarnarekha (gold streak) river have some alluvial gold.

6 KERALA:

The river terraces along the Punna Puzha and the Chaliyar Puzha have some alluvial gold.



HUTTI GOLD MINES

INDIAN MINERAL & ENERGY RESOURCES

NON-FERROUS MINERAL

4

SILVER

1 It is used in chemicals, electroplating, photography and for colouring glass, etc.

2 Silver is usually found combined with other metals in its crude form and in various ores such as argentite.



3 India is not a major producer of silver in the world.

4 **Zawar mines in Udaipur district** of Rajasthan is the major producer of silver.

5 The **Tundoo Lead Smelter in Dhanbad** district of Jharkhand is another major silver producer.

6 Some silver is produced by **Kolar Gold Fields and Hutti gold mines**.

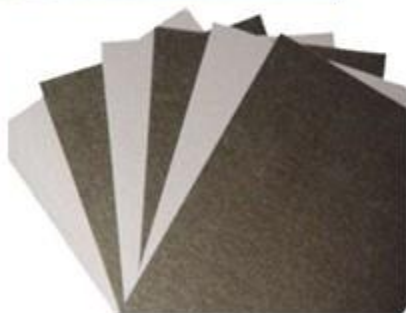
7 The **Hindustan Copper Ltd. in Singhbhum district of Jharkhand** obtains silver from copper slimes.

INDIAN MINERAL & ENERGY RESOURCES

NON-METALLIC MINERALS

MICA

Among the non-metallic minerals produced in India, mica is the important one. The other minerals extracted for local consumption are limestone, dolomite and phosphate.



- 1 Mica is mainly used in the **electrical and electronic industries**.
- 2 It can be split into very thin sheets which are tough and flexible.
- 3 Mica in India is produced in **Jharkhand, Andhra Pradesh, Telangana and Rajasthan** followed by Tamil Nadu, West Bengal and Madhya Pradesh.

INDIAN MINERAL & ENERGY RESOURCES

NON-METALLIC MINERALS

MICA

4 JHARKHAND:

In Jharkhand high quality mica is obtained in a belt extending over a distance of about 150 km in length and about 22 km in width in lower Hazaribagh plateau.

5 ANDHRA PRADESH:

Nellore district produces the best quality mica.

6 RAJASTHAN:

Mica belt extends for about 320 kms from Jaipur to Bhilwara and around Udaipur.

7

Mica deposits also occur in

- Mysore and Hasan districts of Karnataka,
- Coimbatore, Tiruchirapalli, Madurai and Kanyakumari in Tamil Nadu,
- Alleppey in Kerala,
- Ratnagiri in Maharashtra,
- Purulia and Bankura in West Bengal.

ENERGY RESOURCES



- 1 Mineral fuels are essential for generation of power, required by agriculture, industry, transport and other sectors of the economy.
- 2 Mineral fuels like coal, petroleum and natural gas (known as fossil fuels), nuclear energy minerals, are the conventional sources of energy.
- 3 These conventional sources are exhaustible resources.

INDIAN MINERAL & ENERGY RESOURCES

CONVENTIONAL ENERGY SOURCES

NUCLEAR ENERGY RESOURCES

1 Nuclear energy has emerged as a viable source in recent times.

2 Important minerals used for the generation of nuclear energy are uranium and thorium.



A handful of uranium pellets is more powerful than a mountain of coal

3 URANIUM:

- Uranium deposits occur in the **Dharwar** rocks.
- Geographically, uranium ores are known to occur in several locations along the **Singhbhum** Copper belt.
- It is also found in
 - ✓ Udaipur, Alwar and Jhunjhunu districts of Rajasthan,
 - ✓ Durg district of Chhattisgarh,
 - ✓ Bhandara district of Maharashtra and
 - ✓ Kullu district of Himachal Pradesh.

CONVENTIONAL ENERGY SOURCES

India is making nuclear power one of its key policy initiatives, with plans to build 48 new reactors and boost output to 63,000 megawatts by 2032 – an almost 14-fold increase on current levels. The country's existing 20 nuclear reactors generate about 4,700 megawatts



NUCLEAR ENERGY RESOURCES

4 THORIUM:

- Thorium is mainly obtained from monazite and ilmenite in the beach sands along the coast of Kerala and Tamil Nadu.
- World's richest monazite deposits occur in Palakkad and Kollam districts of Kerala, near Vishakhapatnam in Andhra Pradesh and Mahanadi river delta in Odisha.

5

Some of the important nuclear power projects are Tarapur (Maharashtra), Rawatbhata near Kota (Rajasthan), Kalpakkam (Tamil Nadu), Narora (Uttar Pradesh), Kaiga (Karnataka) and Kakrapar (Gujarat).

INDIAN MINERAL & ENERGY RESOURCES

NUCLEAR ENERGY RESOURCES

MONAZITE SANDS AND THE PRESENCE OF THORIUM

- 1 Monazite is a reddish-brown phosphate mineral containing rare earth metals.
- 2 Thorium is estimated to be about 3 to 4 times more abundant than uranium in the Earth's crust, and is chiefly refined from monazite sands.
- 3 According to the Atomic Minerals Directorate for Exploration and Research (AMD), a constituent Unit of the Department of Atomic Energy (DAE), India has 10.70 million tonnes of Monazite which contains 9,63,000 tonnes of Thorium Oxide (ThO_2).

WHAT ARE THE CHIEF ADVANTAGES OF THORIUM?

- 1 Proliferation is not easy. Weapons-grade fissionable material (U-233) is harder to retrieve safely from a thorium reactor. Thus, thorium reactors are significantly more proliferation-resistant than present reactors.
- 2 For India, it offers the added benefit that it can enter the export market as India has the largest reserves of thorium.

INDIAN MINERAL & ENERGY RESOURCES

NON-CONVENTIONAL ENERGY SOURCES

WIND ENERGY

- 1 Wind energy is absolutely **pollution free, inexhaustible** source of energy.



- 2 The mechanism of energy conversion from blowing wind is simple.
- 3 The kinetic energy of wind, through turbines, is converted into electrical energy.
- 4 The permanent wind systems such as the trade winds, westerlies and seasonal wind like monsoon have been used as source of energy.
- 5 Besides these, local winds, land and sea breezes can also be used to produce electricity.

INDIAN MINERAL & ENERGY RESOURCES

WIND ENERGY

WHAT ABOUT INDIA'S STATUS ON WIND ENERGY?

- 1 At present, most of the installed wind capacity is located in the **south, north, and west**. India's east and northeast do not have any grid-connected wind power plants.
- 2 **South India** has the maximum grid-connected installed wind capacity followed by west and north.
- 3 **Tamil Nadu** is the leading producer of wind energy followed by **Maharashtra, Gujarat, and Rajasthan**.
- 4 Around **75 %** of India's wind power generation occurs between the months of **May and September** i.e., in the **southwest monsoon season**.
- 5 Until the end of 2017, around 33 GW of wind power has been installed, according to the Ministry of New and Renewable Energy.



INDIAN MINERAL & ENERGY RESOURCES

WIND ENERGY

PRACTICE QUESTION

Discuss the benefits and limitations of offshore wind energy generation, in relation to onshore wind energy generation. Also discuss steps taken by Indian government to harness offshore wind energy. **(250 words)**



WIND ENERGY



WHAT DO YOU MEAN BY OFFSHORE WIND POWER?

- 1 Offshore wind power or offshore wind energy is the **use of wind farms constructed in bodies of water**, usually in the ocean on the continental shelf, to harvest wind energy to generate electricity.
- 2 Higher wind speeds are available offshore compared to on land, so offshore wind power's electricity generation is **higher per amount of capacity installed**.

WIND ENERGY

WHY THIS QUESTION WAS ASKED?

- 1 Recently, in 2018, India had called for **expression-of-interest** for its first **offshore wind energy project**.
- 2 In 2015, India announced an ambitious goal of increasing renewable power capacity to **175 gigawatts (GW) by 2022**, with 100 GW of solar, 60 GW of wind, 10 GW of bioenergy and 5 GW of small hydro.

QUICK FACT

Chennai-based **National Institute Of Wind Energy (NIWE)** is the nodal agency to carry out the necessary studies and surveys before final bidding and will act as a single window for facilitating necessary clearances required for development of offshore wind project in India.



नीवे NIWE

WIND ENERGY

LET US TAKE A LOOK AT THE PIB ARTICLE

- MNRE calls for 'Expression of Interest' for first offshore wind energy project of 1000 MW capacity
- Plans to install at least 5 GW of offshore wind capacity by 2022

1 The National Institute of Wind Energy (NIWE) an autonomous body under the Ministry of New and Renewable Energy (MNRE) has called for 'Expression of Interest' (Eoi) for the first offshore wind energy project of India.

2 The global Eoi is intended to shortlist prospective offshore wind energy developers for a **1000 MW offshore wind energy project in Gulf of Khambhat, off the coast of Gujarat.**

3 The proposed area is located 23-40 km seaward side from **Pipavav port**. MNRE plans to install at least 5 GW of offshore wind capacity by 2022.

4 As per the **National Offshore Wind Energy Policy** notified on October, 2015, National Institute of Wind Energy (NIWE), Chennai is the nodal agency for development of offshore wind project in India.

INDIAN MINERAL & ENERGY RESOURCES

WIND ENERGY

LET US TAKE A LOOK AT THE PIB ARTICLE
ON OFFSHORE WIND TURBINES....

5 At global level, it has been observed that, offshore wind energy while being better than onshore wind in terms of efficiency is also becoming competitive and comparable in terms of tariffs.

6 This would help the country towards attaining energy security and achievement of National Action Plan for Climate Change (NAPCC) targets.

7 *Nodal Ministry*- the Ministry of New & Renewable Energy (MNRE) has been authorized as the Nodal Ministry for use of offshore areas within the Exclusive Economic Zone (EEZ) of the country.

Nodal agency- the National Institute of Wind Energy (NIWE).

INDIAN MINERAL & ENERGY RESOURCES

WIND ENERGY

PRACTICE QUESTION

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OFFSHORE VS ONSHORE WIND ENERGY

PROS / ADV.



- Offshore Winds are faster than on land and hence there is more energy production in case of Offshore Wind Mills.
- Since the Offshore wind mills are not near any population development they can be made of big size to produce more energy as compared to onshore where size is limited.
- The sea offers no geographical hindrance such as mountains, deserts etc in case of installation and servicing of wind mills.

CONS / DISADV.



- Offshore wind mills are expensive and hard to build and maintain as compared to onshore mills. It has harsher conditions of operations like high humidity, salty sprays from the oceans etc.
- The offshore wind mills may have an adverse impact on marine animals.
- If built near coastline they can disrupt the natural view of the coastline which may impact tourism industry.

WIND ENERGY

STEPS TAKEN BY INDIA TO HARNESS OFFSHORE WIND ENERGY

- 1 Union Government has initiated **National Offshore Wind Energy Policy** in 2015 to produce wind energy and reduce carbon emissions.
- 2 Setting up of a **1000 MW offshore plant in Gulf of Khambat, Gujarat.**

WAY FORWARD

- 1 Since the cost of investment is huge in such cases, the Government needs to **subsidise the sector to promote investment opportunities.**
- 2 Proper surveys and studies before final decision on the Project would provide an idea of challenges possessed on implementing the project.

India is determined to fulfill opportunities under Paris Accord and SDG 2030 Goals for Renewable Energy. The need of hour is increasing our energy generation from Renewable sources of energy to reduce Carbon Emissions and Provide Healthy and Better Future.

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SINCERE THANKS
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